



NORTH SYDNEY BOYS HIGH SCHOOL

2013 HSC ASSESSMENT TASK 1

Mathematics

General Instructions

Section A contains 5 multiple choice questions to be answered on the booklet provided.

Reading Time – 5 minutes
Working Time – 50 minutes
Total Time – 55 minutes

Section B is to be answered in the booklet provided, showing ALL necessary working.

- Write using blue or black pen
- Board approved calculators may be used
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher: Please tick

- ☐ Mr Lowe
- ☐ Mr Berry
- ☐ Mr Lin
- ☐ Mr Lucas
- ☐ Mr Lam
- ☐ Mr Fletcher
- ☐ Ms Ziazaris

Student Number: _____

(To be used by the exam markers only.)

Question No	1-5	6	7	8	9	10	Total	Total %
Mark	5	13	10	10	8	5	51	100

8 marks

Total: 49 marks

SECTION A

Question 1

What is the value of $\frac{dy}{dx}$ if $y = 2\sqrt{x}$?

- (A) $\frac{dy}{dx} = \frac{1}{\sqrt{x}}$
- (B) $\frac{dy}{dx} = \frac{2}{\sqrt{x}}$
- (C) $\frac{dy}{dx} = \frac{\sqrt{x}}{2}$
- (D) $\frac{dy}{dx} = 2$

Question 2

What is the gradient of the curve $y = x^2 - x - 6$ at $(6, 24)$?

- (A) 11
- (B) 12
- (C) 23
- (D) 24

Question 3

What values of x is the curve $f(x) = 2x^3 + x^2$ concave down?

- (A) $x < -\frac{1}{6}$
- (B) $x > -\frac{1}{6}$
- (C) $x < -6$
- (D) $x > 6$

Question 4

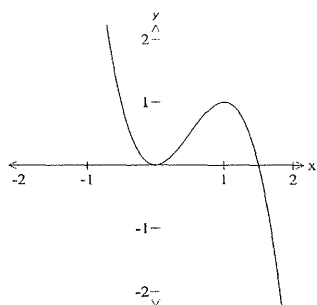
In the first week of the snow season 5 cm of snow falls. In each following week the snowfalls increase by 2 cm, so in the second week there is 7 cm, in the third week there is 9 cm. How much snow falls in the 11th week?

- (A) 23 cm
- (B) 25 cm
- (C) 27 cm
- (D) 29 cm

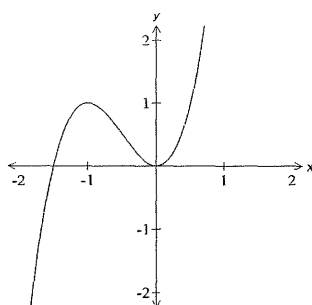
Question 5

Which of the following is the graph of $f(x) = 2x^3 - 3x^2$?

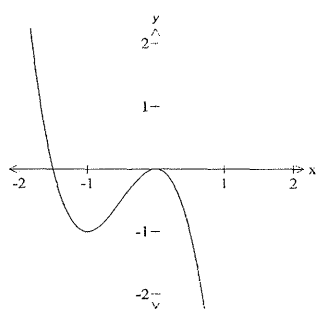
(A)



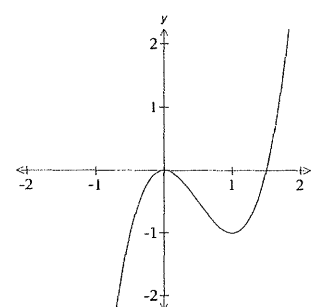
(B)



(C)



(D)



Section B

Question 6 (13 Marks)

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Marks

(a) Differentiate with respect to x :

(i) $1 + 2x - 4x^2 - x^3$ 1

(ii) $\sqrt{1 - x^2}$ 1

(iii) $(x - 1)^5(3x^2 + 1)$ 3

(iv) $\frac{x + 4}{x + 5}$ 2

(b) Find the limit of the following:

(i) $\lim_{x \rightarrow 4} \frac{x}{x + 4}$ 1

(ii) $\lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{x + 4}$ 1

(iii) $\lim_{x \rightarrow \infty} \frac{3x^2 - x + 1}{x^2 - 1}$ 1

(c) Find the coordinates of the point(s) on the curve $y = 3x^2 + 4x + 2$ where the tangent is perpendicular to the line $x + 2y - 10 = 0$ 3

Question 7 (10 Marks)

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Marks

(a) An arithmetic series has a seventh term of 8 and the common difference is -4

(i) What is the first term a ? 1

(ii) Calculate the sum of the first 17 terms. 2

(b) Evaluate $\sum_{k=2}^5 (-1)^k \left(\frac{1}{k}\right)$ 2

(c) The third term of a geometric series is $\frac{3}{4}$ and the seventh term is 12.

(i) Find the 14th term of the series. 3

(ii) Find the sum of the first 14 terms. 2

Question 8 (10 Marks)

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Marks

Consider the curve given by $y = \frac{1}{4}x^4 - x^3$

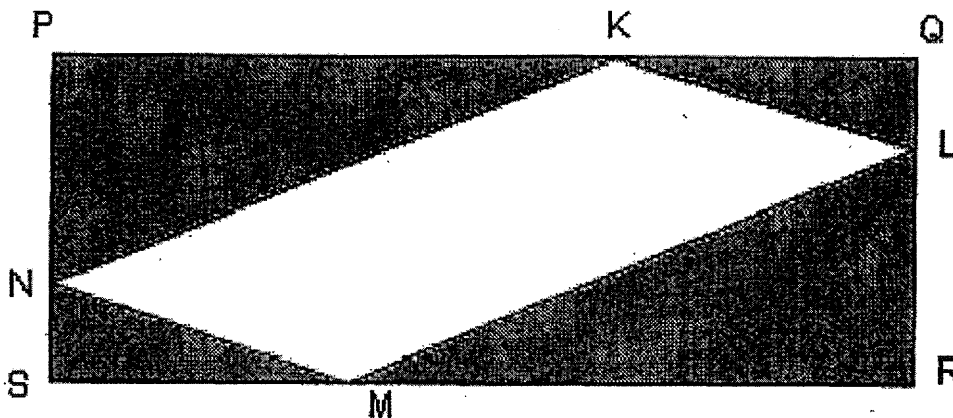
- (i) Find any turning points and determine their nature. 4
- (ii) Find any points of inflexion. 3
- (iii) Sketch the curve for $-1 \leq x \leq 5$, indicating where the curve crosses the x axis. 2
- (iv) For what values of x is the curve concave down? 1

Question 9 (8 Marks)

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Marks

- (a) In the diagram below, PQRS is a rectangle with PQ = 40cm and SP = 10 cm. The shaded portions are cut away, leaving away a parallelogram KLMN. QL = SN = x cm and QK = SM = $4x$ cm.



- (i) Show that the area of the parallelogram KLMN is given by $A = 80x - 8x^2$ 2
 - (ii) What is the domain of x ? 1
 - (iii) Find the maximum area of the parallelogram KLMN 2
- (b) Sketch the graph of the function that satisfies the given conditions: 3
- $\lim_{x \rightarrow 3} f(x) = -\infty, \quad f''(x) < 0 \text{ if } x \neq 3, \quad f'(0) = 0$
- $f'(x) > 0 \text{ if } x < 0 \text{ or } x > 3, \quad f'(x) < 0 \text{ if } 0 < x < 3$

Question 10 (5 Marks)

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Marks A_n and B_n are two series given by

$$A_n = 1^2 + 5^2 + 9^2 + 13^2 + \dots + (4n - 3)^2$$

$$B_n = 3^2 + 7^2 + 11^2 + 15^2 + \dots$$

- | | | |
|-------|--|---|
| (i) | Find the n th term of B_n | 1 |
| (ii) | If $S_{2n} = A_n - B_n$ Prove that $S_{2n} = -8n^2$ | 2 |
| (iii) | Hence or otherwise evaluate
$101^2 - 103^2 + 105^2 - 107^2 + \dots + 2001^2 - 2003^2$ | 2 |

END OF EXAMINATION

MC Answers: 1. A 2. A 3. A 4. B 5. D.

Question 6

a) i) $2 - 8x - 30x^2$ (1 mark)

ii) $f(x) = \sqrt{1-x^2}$
 $= (1-x^2)^{\frac{1}{2}}$

$$f'(x) = \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \cdot -2x$$
$$= -\frac{x}{\sqrt{1-x^2}} \quad (1 \text{ mark})$$

iii) $f(x) = (x-1)^5 (3x^2+1)$

$$f'(x) = (x-1)^5 \cdot \frac{d}{dx}(3x^2+1) + (3x^2+1) \cdot \frac{d}{dx}(x-1)^5$$

$$= (x-1)^5 \cdot 6x + (3x^2+1) \cdot 5(x-1)^4 (*)$$

$$= (x-1)^4 [(x-1) \cdot 6x + (3x^2+1) \cdot 5]$$

$$= (x-1)^4 [6x^2 - 6x + 15x^2 + 5]$$

$$= (x-1)^4 [21x^2 - 6x + 5] \quad (3 \text{ marks})$$

iv) $f(x) = \frac{x+4}{x+5}$

$$f'(x) = \frac{(x+5) \cdot \frac{d}{dx}(x+4) - (x+4) \cdot \frac{d}{dx}(x+5)}{(x+5)^2}$$

$$= \frac{(x+5) - (x+4)}{(x+5)^2}$$

$$= \frac{1}{(x+5)^2}$$

(2 marks)

$$b) \quad (i) \quad \lim_{x \rightarrow 4} \frac{x}{x+4}$$

$$= \frac{4}{4+4}$$

$$= \frac{1}{2}$$

(1 mark)

$$(ii) \quad \lim_{x \rightarrow -4} \frac{(x+4)(x-1)}{(x+4)}$$

$$= \lim_{x \rightarrow -4} x-1$$

$$= -5$$

(1 mark)

$$(iii) \quad \lim_{x \rightarrow \infty} \frac{3x^2 - x + 1}{x^2 - 1}$$

$$= \lim_{x \rightarrow \infty} \frac{3 - \frac{1}{x} + \frac{1}{x^2}}{1 - \frac{1}{x^2}}$$

$$= 3$$

(1 mark)

$$c) \quad y = 3x^2 + 4x + 2$$

gradient
of
tangent:

$$\frac{dy}{dx} = 6x + 4$$

$$x + 2y - 10 = 0 \Rightarrow y = -\frac{x}{2} + 10$$

gradient perpendicular to line has gradient = 2

$$\frac{dy}{dx} = 2 \Rightarrow$$

$$6x + 4 = 2$$

$$6x = -2$$

$$x = -\frac{1}{3}$$

$$\therefore y = 3\left(-\frac{1}{3}\right)^2 + 4\left(-\frac{1}{3}\right) + 2$$

$$= \frac{1}{3} - \frac{4}{3} + \frac{6}{3}$$

$$= 1$$

$\therefore$ co ordinate $\left(-\frac{1}{3}, 1\right)$ (3 marks)

Question 7

a) (i) $T_n = a + (n-1)d$

$T_7 = 8 \quad d = -4$

$\therefore T_7 = a + 6 \cdot (-4)$

$\therefore a - 24 = 8$

$a = 32$

(1 mark)

(ii) $S_{17} = \frac{n}{2} [2a + (n-1)d]$

$= \frac{17}{2} [64 + (16 \times -4)]$

$= 0$

(2 marks)

b)

$\sum_{k=2}^5 (-1)^k \left(\frac{1}{k}\right)$

$= (-1)^2 \left(\frac{1}{2}\right) + (-1)^3 \left(\frac{1}{3}\right) + (-1)^4 \left(\frac{1}{4}\right) + (-1)^5 \left(\frac{1}{5}\right)$

$= \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5}$

$= \frac{13}{60}$

(2 marks)

c)

$T_n = ar^{n-1}$

$T_3 = \frac{3}{4}$

$T_7 = 12$

$ar^2 = \frac{3}{4} \text{ --- (1)} \quad ar^6 = 12 \text{ --- (2)}$

$\therefore \textcircled{2} \div \textcircled{1}$

$r^4 = 16$

$\therefore r = \pm 2$

sub into ①

$\therefore 4a = \frac{3}{4}$

$\therefore a = \frac{3}{16}$

$\therefore T_{14} = ar^{13}$

$= \frac{3}{16} (-2)^{13} \text{ or } \frac{3}{16} (2)^{13}$

$= -1536 \text{ or } 1536$

(3 marks)

c) ii) $S_{14} = \frac{a(r^{14}-1)}{r-1}$

$$= \frac{\frac{3}{16} [(-2)^{14} - 1]}{-3}$$

$$= \frac{16383}{16}$$

or $\frac{\frac{3}{16} [(2)^{14} - 1]}{1}$

$$= \frac{49149}{16}$$

(2 marks)

Question 8

$$y = \frac{1}{4}x^4 - x^3$$

$$i) \frac{dy}{dx} = x^3 - 3x^2$$

stationary points occur when $\frac{dy}{dx} = 0$

$$\Rightarrow x^3 - 3x^2 = 0$$

$$x^2(x-3) = 0$$

$$\therefore x = 0 \text{ or } x = 3$$

$$\frac{d^2y}{dx^2} = 3x^2 - 6x$$

$$\text{at } x = 0$$

$$\frac{d^2y}{dx^2} = 0$$

$$\text{at } x = 3$$

$$\frac{d^2y}{dx^2} = 9 > 0$$

possible point of inflexion.

$\therefore$ minimum at $x = 3$

$\therefore$ minimum turning point at $(3, -\frac{27}{4})$

x	-1	0	1
$\frac{d^2y}{dx^2}$	9	0	-3
	∪	•	∩

$\therefore$ point of inflexion at $x = 0$

$\therefore$ horizontal point of inflexion at $x = 0$

i.e. $(0, 0)$

(4 marks)

ii) Points of inflexion occurs when $\frac{d^2y}{dx^2} = 0$

From (i) $\frac{d^2y}{dx^2} = 3x^2 - 6x$

$$\therefore 3x^2 - 6x = 0$$

$$3x(x-2) = 0$$

$$\therefore x = 0 \text{ or } x = 2$$

From (i) $x = 0$ is a horizontal point of inflexion.
testing for concavity.

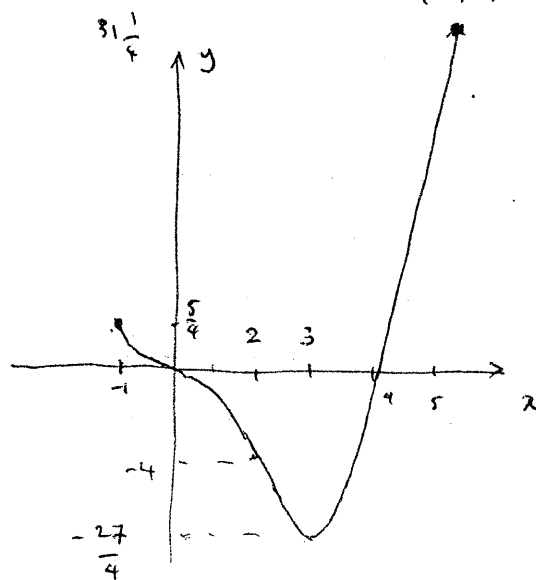
x	1	2	3
$\frac{d^2y}{dx^2}$	-3	0	9

$\curvearrowright$ $\curvearrowleft$

$\therefore$ change in concavity

$\therefore$ point of inflexion at $x = 2$.

$\therefore$ points of inflexion are at (3 marks)
 $(0, 0)$ and $(2, -4)$



iii)

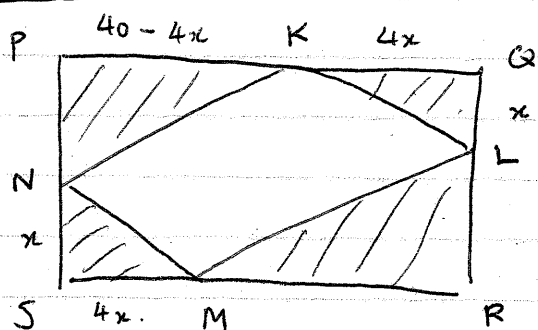
iv)

$$0 < x < 2.$$

(1 mark)

Question 9

a) i)



$$\begin{aligned}
 \text{Area of parallelogram} &= 400 - 2 \left(\frac{1}{2} \times (10-x)(40-4x) \right) - 2 \left(\frac{1}{2} \times 4x \times x \right) \\
 &= 400 - [400 - 80x + 4x^2] - 4x^2 \\
 &= 80x - 8x^2 \quad (2 \text{ marks})
 \end{aligned}$$

ii) $0 < x < 10$ (1 mark)

iii) $A = 80x - 8x^2$

$$\frac{dA}{dx} = 80 - 16x$$

$$\frac{dA}{dx} = 0 \Rightarrow 80 - 16x = 0$$

$$16x = 80$$

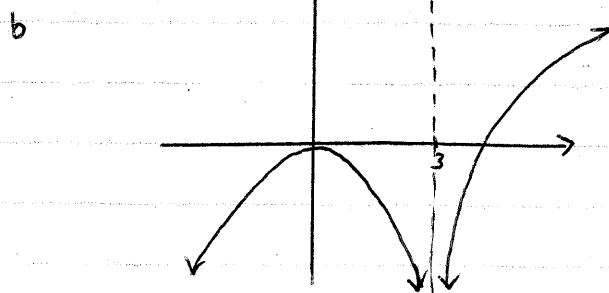
$$x = 5$$

$$\frac{d^2A}{dx^2} = -16 < 0 \text{ for all values}$$

$\therefore$ maximum occurs at $x = 5$

$$\therefore \text{maximum area} = (80 \times 5) - (8 \times 5^2)$$

$$= 200 \text{ cm}^2 \quad (2 \text{ marks})$$



(3 marks)

Question 10

$$\begin{aligned} \text{i)} \quad & (3 + (n-1)4)^2 \\ &= (3 + 4n - 4)^2 \\ &= (4n - 1)^2 \quad (1 \text{ mark}) \end{aligned}$$

$$\begin{aligned} \text{ii)} \quad S_{2n} &= A_n - B_n \\ &= (1^2 + 5^2 + 9^2 + 13^2 + \dots + (4n-3)^2) - (3^2 + 7^2 + 11^2 + 15^2 + \dots + (4n-1)^2) \\ &= (1^2 - 3^2) + (5^2 - 7^2) + (9^2 - 11^2) + \dots + [(4n-3)^2 - (4n-1)^2] \\ &= -8 - 24 - 40 + \dots + (-16n + 8) \end{aligned}$$

this is a AP, $a = -8$, $n = -16$

$$\begin{aligned} \therefore S_{2n} &= \frac{n}{2} [a + l] \\ &= \frac{n}{2} [-8 + (-16n + 8)] \\ &= -8n^2 \quad (2 \text{ marks}) \end{aligned}$$

$$\begin{aligned} \text{iii)} \quad & 101^2 - 103^2 + 105^2 - 107^2 + \dots + 2001^2 - 2003^2 \\ &= S_{1000} - S_{50} \\ &= -8(500)^2 - (-8 \times 25^2) \\ &= -2003008 \quad (2 \text{ marks}) \end{aligned}$$

Alternatively they can do

$(101^2 - 103^2)$ as first term of AP

$(2001^2 - 2003^2)$ as last term and

find the number of terms in between